

Population & Migration Predictions in Puerto Rico

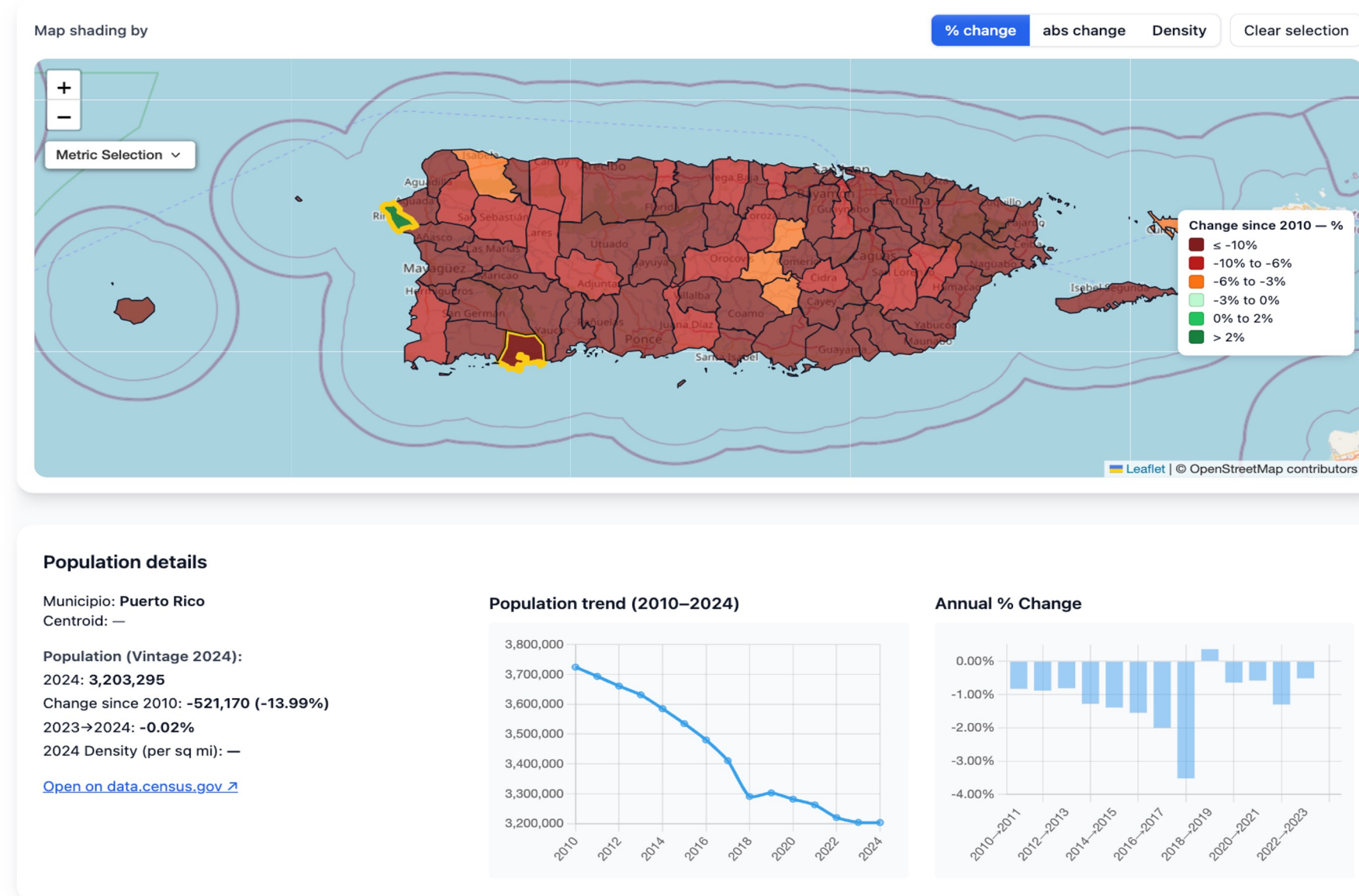
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Abstract

Puerto Rico has experienced sustained population decline driven by natural disasters, social conditions, and disproportionate resources allocation. Understanding how these forces interact is critical for:

- Predicting future population trends
- Identifying vulnerable regions
- Informing government policy responses

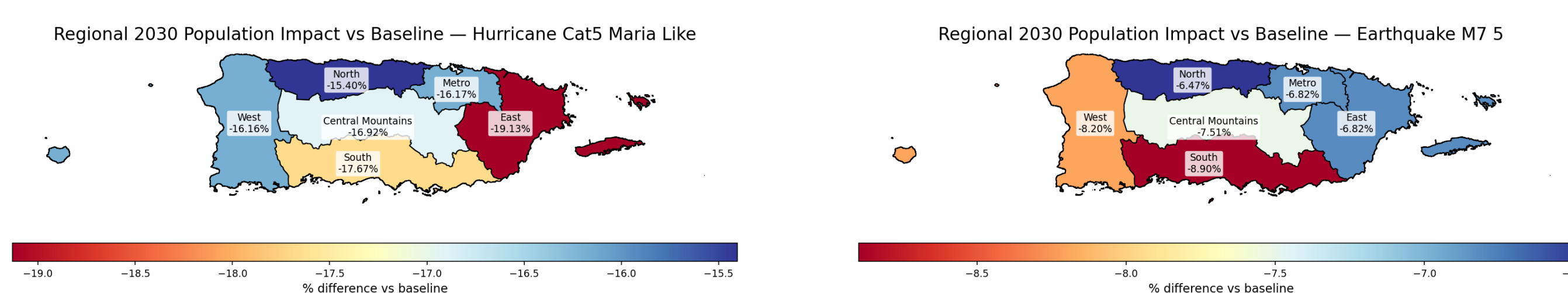
This project applies machine learning and scenario analysis to model regional-level population change and evaluate how structural dynamics dominate long-term population change, with natural disasters acting as amplifiers.



Background

- Puerto Rico has seen persistent population loss since ~2010
- Decline is uneven across regions, with the southern region being more affected
- Hurricanes and earthquakes introduce long-term shocks and volatility
- Long-term dynamics are driven by:
 - Natural disasters
 - Migration patterns
 - Economic opportunity
 - Safety and social vulnerability indicators (SVIs)

Key challenge: Separating short-term shocks vs long-term structural trends.

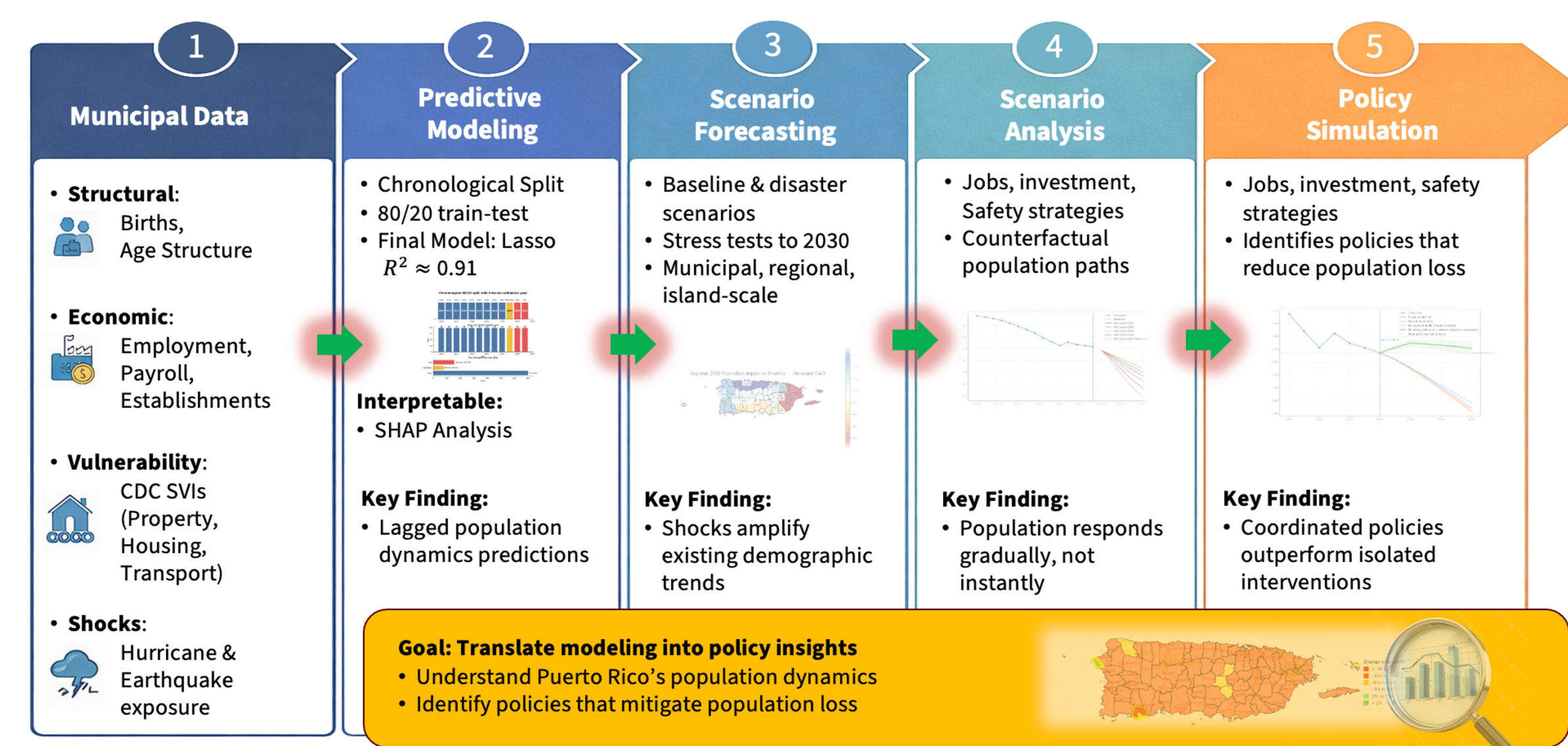


Objectives

- Identify key drivers of population change
- Evaluate the role of key drivers:
 - Natural disasters (hurricanes & earthquakes)
 - Socioeconomic conditions
 - Government migration strategies
- Develop scenario forecasts through 2030
- Assess how policy-relevant factors influence outcomes

Methods

- **Exploratory Data Analysis:**
 - Municipal-level (2010–2023)
 - Economic, safety, demographic, vulnerability, disaster features
 - Reference: U.S. Census Bureau. American Community Survey
- **Split:**
 - Train ≤ 2018
 - Validation = 2019
 - Test ≥ 2020
 - (chronological $\rightarrow$ no data leakage)
- **Final Model:**
 - Lasso ($R^2 \approx 0.91$) selected for strong out-of-sample performance and interpretability
- **Features:**
 - Hurricane and earthquake exposure
 - Crime rates
 - Income & establishment growth
 - Social vulnerability indicators
 - Natural change (births–deaths)
- **Results Interpretation:**
 - Predictive population graphical representations



Results & Discussion

Key findings:

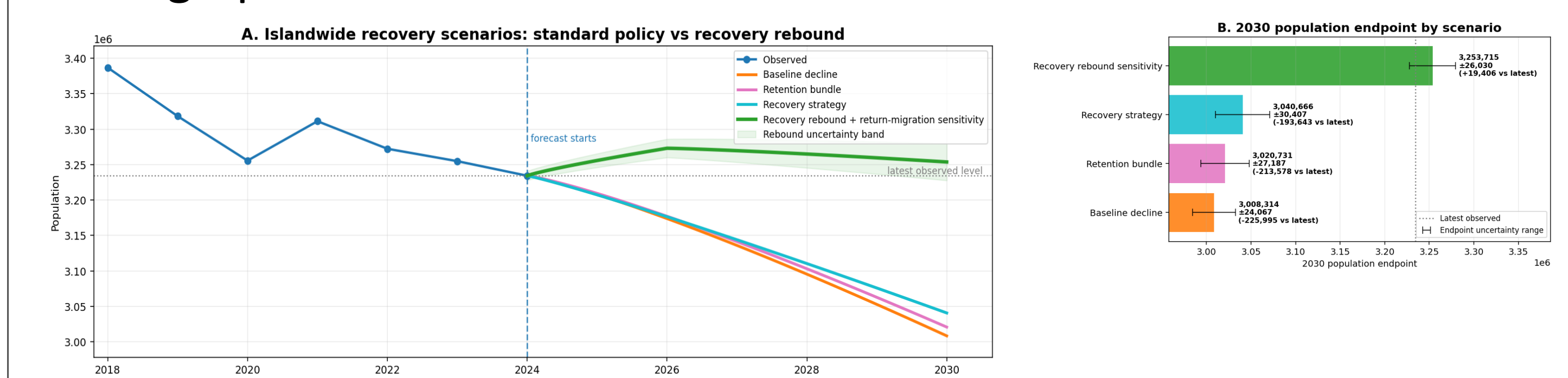
- Population decline follows a persistent trajectory across the island
 - Without strong intervention, existing trends continue over time
- Disasters are strong direct predictors
- Strong regional disparities
 - Vulnerability and exposure vary significantly
- Model performance:
 - Lasso achieves strong out-of-sample results
 - Stable across data split
- Scenario insight:
 - Shocks amplify existing migration trends
 - Policy bundles can slow decline, but population increase requires strong, coordinated recovery that supports return migration and improves structural conditions

Interpretation:

Population decline is driven primarily by persistent structural dynamics, with disasters acting as accelerators rather than root causes

Main takeaway:

Population decline in Puerto Rico is persistence-driven; reversing it requires a coordinated recovery strong enough to overcome demographic momentum



Scenario	What changes in the model?	Magnitude	What it means
Baseline decline	No policy-feature adjustments	Reference path	Continues stylized decline path
Retention bundle	Jobs + safety + vulnerability + demographic support changes	20% job/safety; 15% vulnerability; +0.35 natural change; +0.28 pop-momentum proxy	Standard coordinated policy package; slows decline
Recovery bundle (model-implied)	Stronger version of retention bundle	30-35% jobs; 22-25% safety/vulnerability; +0.45 natural change; +0.75 pop-momentum proxy	Model-implied recovery improves trajectory but still declines
Recovery rebound + return migration	Uses recovery bundle, then adds explicit rebound layer	+1.2% temporary rebound peak; +0.6% 2030 floor above latest observed	Sensitivity case: recovery momentum needed for stabilization/growth

References

- U.S. Census Bureau. American Community Survey (ACS)
- U.S. Census Bureau. Population Estimates Program (PEP) Puerto Rico Municipal Population Estimates, Vintage 2024
- NOAA National Hurricane Center
- USGS (United States Geological Survey)